

Department of Computer Science and Engineering
Jashore University of Science & Technology
3rd Year 1st Semester (Session: 2021-22)

Course Code: CSE-3107

Course Title: Software Engineering

Total Time: 3.00 hours

Total Marks: 72.0

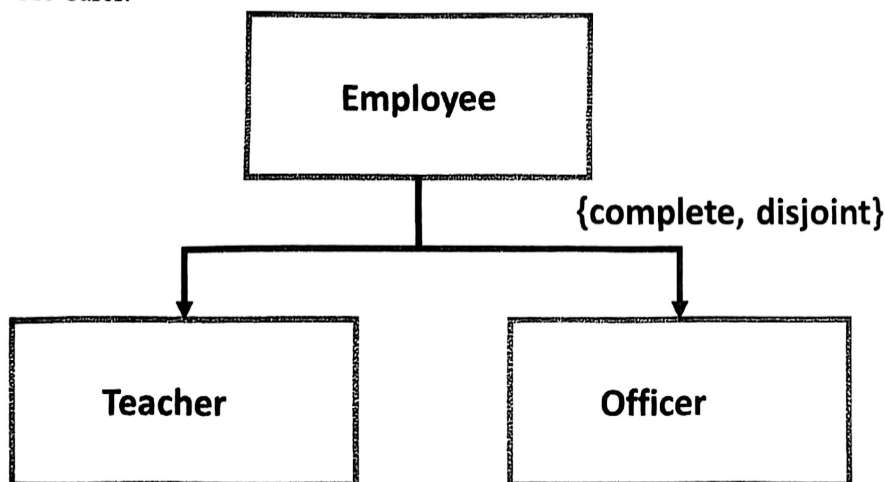
(Note: Answer any six set of questions from the following eight set of questions)

- ✓ 1 (a) Briefly discuss the software engineering core principles that the development team should keep in mind at the start of a project. 4
- (b) What are pull requests and software releases in the version control system (i.e., GitHub)? 2
- (c) Define Software Engineering. What are the activities in generic process framework for software engineering? Describe. 6

- 2 (a) Describe different Perspective Process Models in Software Engineering. How Incremental Process Model is different from Prototyping? 9
- (b) In which scenario Waterfall Model is better than Incremental Model? Explain. 3

- ✓ 3 (a) Draw a Use Case Diagram for "Room Allocation" for a Hotel Management System where the customer asks for a key for a room which he/she has already booked. After security clearance and checking availability of appropriate kind of room, the front desk officer asks for the payment. After payment is done on site (Coupon Codes can also be used for discounts), the front desk officer dispatches the room key to the customer. Figure out available classes and their relationships from the Use Cases. 8

- 7 (b) 4



Write classes with appropriate properties and getRemuneration() method from the above-mentioned diagram.

- ✓ 4 (a) Consider a scenario where a customer withdraws cash from ATM Machine. His/her account must contain at least 500 Tk. after a successful withdrawal. If he/she intends to withdraw below that amount, he/she must accept that a fine will be deducted later on. Three consecutive wrong pins will not allow any further attempts for the next 24 hours. Stolen card will be confiscated at once. Balance will be displayed and the card will be dispatched in applicable scenario. 6

Draw an Interaction Diagram for above-mentioned scenario.

- (b) Draw an Activity Diagram using swim lanes for above-mentioned scenario. 6

- ✓ 5 (a) We have a program segment like below: 9
- variable x, y, z assignment.
While z<10

```

if x>5 then
  if y<2 then
    p=abc;
  elseif y>2 then
    q=xyz;
  endif
else
  m=ijk;
  n=lmnop;
endif
z=z+1;
x=x+1;
y=y+1;
end while
print variables;

```

- ✓ Prepare a Flow-graph for this program and identify all independent program paths by identifying cyclomatic complexity. Now, derive test cases for checking all the program paths whether they work as per specification or not. Give emphasis on Boundaries.
- 8 (b) What are the advantages of employing the Pipe-Filter Architectural pattern? When it is used? 3
- 6 (a) Describe Software Testing Strategies in detail. 9
- (b) Suppose you are configuring a world-renowned ERP software (like SAP) for a large company. What type of testing would be the most crucial in this project in your opinion? 3
- 7 (a) Explain the following code smells with proper example: (i) Improper Inheritance, (ii) Divergent Change, (iii) Improper error handling. 6
- (b) Suppose you have gone to KFC with a relative and have ordered a menu. Draw a State Diagram for the Order Class. In which testing State Diagrams can be helpful? Explain. 6
- 8 (a) What is a Project? What are the phases of a project? 4
- (b) Describe Programs, Subprojects, and Portfolios. 4
- (c) Describe Score Matrices to evaluate a project. 4